THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **Chemical Reactions and Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Rusting of iron is a chemical change. Reason (R): A new substance with different properties (iron oxide) is formed. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. 12 g of carbon burns completely in 32 g of oxygen. The mass of carbon dioxide formed is: 1

(a) 20 g **(b)** 28 g **(c)** 12 g **(d)** 44 g

3. A metal X turns black on heating in air. The black product, on passing hydrogen over it, turns reddish brown. X is: 1

(a) zinc **(b)** copper **(c)** iron **(d)** magnesium

4. Assertion (A): Decomposition of calcium carbonate is an endothermic reaction. Reason (R): Heat energy is absorbed to break the bonds in the compound. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Mg burns in oxygen: 24 g Mg gives 40 g MgO. Mass of oxygen used is: 1

(a) 16 g **(b)** 64 g **(c)** 24 g **(d)** 40 g

6. A student places metal M in blue copper sulphate solution. The blue colour fades to colourless and a reddish-brown solid forms. M could be: 1

(a) silver **(b)** mercury **(c)** gold **(d)** zinc

7. Assertion (A): Chips packets are flushed with nitrogen. Reason (R): Nitrogen is a reactive gas that speeds up oxidation of oil. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. When 4 g of hydrogen completely reacts with 32 g of oxygen, the mass of water formed is: 1

(a) 34 g **(b)** 36 g **(c)** 40 g **(d)** 28 g

9. **Assertion (A):** Burning of magnesium ribbon in air is a combination reaction. 1
Reason (R): In a combination reaction two or more substances combine to form a single product.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A white precipitate is formed when barium chloride solution is mixed with sodium sulphate solution. 1

Reason (R): The reaction between barium chloride and sodium sulphate is a double displacement reaction.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. 12 g of carbon burns completely in 32 g of oxygen. Find the mass of carbon dioxide formed and name the law applied. 2

12. Why is the mass of the residue less than the mass of lead nitrate taken when it is heated in an open tube? 2

13. Write the balanced equation for the thermal decomposition of zinc carbonate and name the gas evolved. 2

14. Why does an iron nail kept for some time in copper sulphate solution show an increase in mass? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Balance $PbO + C \longrightarrow Pb + CO$ and identify the oxidising and reducing agents. 3

16. Green coating forms on copper vessels kept in humid air. Name the coating, give its cause and one prevention. 3

17. Dilute hydrochloric acid is added to zinc granules. Name the gas, state the test for it and write the equation. 3

OR

Balance $Fe_3O_4 + H_2 \longrightarrow Fe + H_2O$ and name the substance oxidised. 3

18. Element X is burnt in air with a dazzling white flame giving a white powder Y. Y dissolved in water turns red litmus blue. Identify X and Y and give the equation. 3

OR

Copper powder is heated in a china dish and then hydrogen is passed over the product. Describe the changes and equations. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a school laboratory, Rohan took about 2 g of green ferrous sulphate crystals in a dry boiling tube and heated it strongly. The crystals first lost water and turned white, then changed to a reddish-brown solid, and a smell like burning sulphur was noticed. 5

(i) What is the colour of the final solid residue?

(ii) Name the type of reaction taking place on strong heating.

(iii) Write the balanced chemical equation for the decomposition of anhydrous ferrous sulphate.

(iv) Name the two gaseous products and state whether the reaction is endothermic or exothermic, with a reason.

20. Zinc and copper are separately treated with dilute sulphuric acid. State the observations and equations, and explain using reactivity. 5

OR

Explain why 5

(i) Zn displaces Cu from CuSO4,

(ii) iron nail gets brown coating,

(iii) AgCl turns grey in sunlight,

(iv) green coat forms on copper. Write equation for

(i) and (iii).

END OF QUESTION PAPER — Total: 40 Marks